14.7 Carolee and I construct a time machine from a wormhole. *Left:* I stay at home in Pasadena with one mouth of the wormhole and hold hands with Carolee through the wormhole. *Right:* Carolee carries the other mouth on a high-speed trip through the Universe. *Inset:* Our hands inside the wormhole.

Carolee departs at 9:00 A.M. on 1 January 2000, as measured by herself, by me, and by everybody else on Earth. Carolee zooms away from Earth at nearly the speed of light for 6 hours as measured by her own time; then she reverses course and zooms back, arriving on the front lawn 12 hours after her departure as measured by her own time.[9] I hold hands with her and watch her through the wormhole throughout the trip, so obviously I agree, *while looking through the wormhole*, that she has returned after just 12 hours, at 9:00 P.M. on 1 January 2000. Looking through the wormhole at 9:00 P.M., I can see not only Carolee; I can also see, behind her, our front lawn and our house.

Then, at 9:01 P.M., I turn and look out the window—and there I see an empty front lawn. The spaceship is not there; Carolee and the other wormhole mouth are not there. Instead, if I had a good enough telescope pointed out the window, I would see Carolee's spaceship flying away from Earth on its outbound journey, a journey that as measured on Earth, *looking through the external Universe*, will require 10 years. [This is the standard "twins paradox"; the high-speed "twin" who goes out and comes back (Carolee) measures a time lapse of only 12 hours, while the "twin" who stays behind on Earth (me) must wait 10 years for the trip to be completed.]

I then go about my daily routine of life. For day after day, month after month, year after year, I carry on with life, waiting—until finally, on 1 January 2010, Carolee returns from her journey and lands on the front lawn. I go out to meet her, and find, as expected, that she has aged just 12 hours, not 10 years. She is sitting there in the spaceship, her hand thrust into the wormhole mouth, holding hands with somebody. I stand behind her, look into the mouth, and see that the person whose hand she holds is myself, 10 years younger, sitting in our living room on 1 January 2000. The wormhole has become a time machine. If I now (on 1 January 2010) climb into the wormhole mouth in the spaceship, I will emerge through the other mouth in our living room on 1 January 2000, and there I will meet my younger self. Similarly, if my younger self climbs into the mouth in the living room on 1 January 2000, he will emerge from the mouth in the spaceship on 1 January